

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250271792

Kind Code

A1

Publication Date

August 28, 2025

Inventor(s)

KATAGIRI; Tatsuya

IMAGE FORMING APPARATUS

Abstract

The image forming apparatus includes: a fixer that heats and fixes a toner image formed on a continuous medium; a collector that collects ultra fine particles generated from an elastic member included in the fixer; an air blower that generates an air current toward a side of the collector; and a hardware processor that varies an output mode of the air blower according to at least one of an apparatus state of the image forming apparatus and/or a printing condition for the image forming apparatus.

Inventors: KATAGIRI; Tatsuya (Tokyo, JP)

Applicant: Konica Minolta, Inc. (Tokyo, JP)

Family ID: 1000008494823

Appl. No.: 19/063567

Filed: February 26, 2025

Foreign Application Priority Data

JP 2024-029010

Feb. 28, 2024

Publication Classification

Int. Cl.: G03G15/20 (20060101); G03G15/00 (20060101); G03G15/01 (20060101); G03G21/20 (20060101)

U.S. Cl.:

CPC G03G15/2025 (20130101); G03G15/0189 (20130101); G03G15/2064 (20130101); G03G15/50 (20130101); G03G15/6591 (20130101); G03G15/6594 (20130101); G03G21/206 (20130101);

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] Japanese Patent Application No. 2024-029010 filed on Feb. 28, 2024, including description, claims, drawings, and abstract the entire disclosure is incorporated herein by reference in its entirety.

BACKGROUND

Technological Field

[0002] The present invention relates to an image forming apparatus that forms an image on a continuous medium.

Description of Related Art

[0003] In a case where a medium to be conveyed in an image forming apparatus is a continuous medium such as continuous paper (for example, roll paper), there is no interval between pieces of papers, unlike the case of flat paper. Since the continuous medium is present all the time in a fixer while image formation is in progress, the heat of the fixer is likely to be absorbed by the continuous medium. For this reason, the temperature of the fixer is set higher in the case of forming an image on a continuous medium than in the case of forming an image on flat paper.

[0004] When the temperature of the fixer is set high, numerous ultra fine particles are generated from an elastic member of the fixer. In a case where the ultra fine particles adhere to a charged electrode of an electric charger, image streaks occur in the conveyance direction, and in a case where the ultra fine particles adhere to a continuous medium, image transferability is affected and image unevenness occurs. That is, the generation of ultra fine particles from the elastic member leads to deterioration of the image quality.

[0005] For this reason, an image forming apparatus capable of removing such ultra fine particles has been developed (see Japanese Patent Application Laid-Open No. 2022-071368). The image forming apparatus described in Japanese Patent Application Laid-Open No. 2022-071368 includes: a filter (referred to as an air blowing filter in Japanese Patent Application Laid-Open No. 2022-071368) that collects ultra fine particles; and an air blowing fan that generates an air current on the side of the filter in the image forming apparatus. The air blowing fan sucks air in the image forming apparatus to the side of the filter. In other words, the air blowing fan sucks ultra fine particles included in the air in the image forming apparatus to the side of the filter.

SUMMARY

[0006] Incidentally, in general, image formation (printing) cannot be interrupted during conveyance in the case of a continuous medium. For this reason, ultra fine particles are continuously generated from an elastic member of a fixer. Further, in the case of a continuous medium, the presence of a continuous medium at the fixer all the time while image formation is in progress reduces the air permeability of the fixer, which may therefore lead to a continuous increase in the amount of ultra fine particles floating around the fixer or inside the image forming apparatus. For this reason, it is necessary to increase the collectability of ultra fine particles by a filter by setting the output (output mode) of an air blowing fan to be high.

[0007] However, when the output of an air blowing fan increases, fan noise increases, it becomes difficult to attempt a noise reduction in the image forming apparatus, power consumption of the air blowing fan increases, and it becomes difficult to attempt energy saving for the image forming apparatus.

[0008] Accordingly, an object of the present invention is to provide an image forming apparatus capable of forming a high-quality image on a continuous medium and attempting a noise reduction and energy saving.

[0009] An aspect of an image forming apparatus according to the present invention is an image

forming apparatus that forms an image on a continuous medium, and the image forming apparatus includes: [0010] a fixer that includes an elastic member, which comes into direct or indirect contact with the continuous medium, and heats and fixes a toner image formed on the continuous medium; [0011] a collector that collects ultra fine particles generated from the elastic member; [0012] an air blower that generates an air current toward a side of the collector; and [0013] a hardware processor that varies an output mode of the air blower according to at least one of an apparatus state of the image forming apparatus and/or a printing condition for the image forming apparatus.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0014] The advantages and features provided by one or more embodiments of the invention will become more fully understood from the detailed description given hereinbelow and the appended drawings which are given by way of illustration only, and thus are not intended as a definition of the limits of the present invention:

[0015] FIG. 1 is a schematic diagram of an overall configuration of an image forming apparatus according to the present embodiment;

[0016] FIG. 2 is an enlarged view of a portion II in FIG. 1;

[0017] FIG. 3 is an enlarged view of a portion III in FIG. 1;

[0018] FIG. 4 is a cross-sectional view taken along a line IV-IV in FIG. 3;

[0019] FIG. 5 is a cross-sectional view taken along a line V-V in FIG. 2;

[0020] FIG. 6 is a control block diagram of the image forming apparatus according to the present embodiment; and

[0021] FIG. 7 is a flowchart for describing setting processing of an output mode of a plurality of first air blowing fans and a plurality of second air blowing fans.

DETAILED DESCRIPTION OF EMBODIMENTS

[0022] Hereinafter, one or more embodiments of the present invention will be described with reference to the drawings. However, the scope of the invention is not limited to the disclosed embodiments.

[0023] Note that, in the specification and claims of the present application, the upstream side refers to the upstream side in the conveyance direction of a continuous medium, and the downstream side refers to the downstream side in the conveyance direction of the continuous medium.

[0024] The configuration of an image forming apparatus **10** according to the present embodiment will be described with reference to FIGS. 1 to 5. FIG. 1 is a schematic diagram illustrating an overall configuration of the image forming apparatus **10** according to the present embodiment. FIG. 2 is an enlarged view of the portion II in FIG. 1. FIG. 3 is an enlarged view of the portion III in FIG. 1. FIG. 4 is a cross-sectional view taken along the line IV-IV in FIG. 3. FIG. 5 is a cross-sectional view taken along the line V-V of FIG. 2.

[0025] As illustrated in FIG. 1, the image forming apparatus **10** is a so-called tandem-type image forming apparatus, and forms (prints) an image on a continuous medium P by an electrophotographic method. As the continuous medium P, for example, continuous paper, tack paper, a polyethylene terephthalate (PET) film, or the like is used.

[0026] The image forming apparatus **10** includes: a printing machine **12** which is the main constituent element thereof; and an unwinding machine **14** that unwinds the continuous medium P to the side of the printing machine **12**, and the unwinding machine **14** is disposed on the right side of the printing machine **12**. The image forming apparatus **10** includes a winding machine **16** that winds the continuous medium P, on which an image has been formed (printed), from the side of the printing machine **12**, and the winding machine **16** is disposed on the left side of the printing machine **12**.

[0027] The unwinding machine **14** includes an unwinding machine main body **18** having a housing shape, and an unwinding shaft **20** that supports the continuous medium P in a roll shape is rotatably provided in the unwinding machine main body **18**. An unwinding motor (not illustrated) for rotating the unwinding shaft **20** is provided at an appropriate position of the unwinding machine main body **18**. A plurality of input roller pairs **22** for inputting the continuous medium P to the side of the printing machine **12** is rotatably provided in the unwinding machine main body **18**. The continuous medium P can be unwound to the side of the printing machine **12** by rotating the unwinding shaft **20** by the driving of the unwinding motor.

[0028] The winding machine **16** includes a winding machine main body **24** having a housing shape, and a winding shaft **26** which supports the continuous medium P, on which an image has been formed, in a roll shape is rotatably provided in the winding machine main body **24**. A winding motor (not illustrated) that rotates the winding shaft **26** is provided at an appropriate position of the winding machine main body **24**. A plurality of output roller pairs **28** for outputting the continuous medium P, on which an image has been formed, from the side of the printing machine **12** is rotatably provided in the winding machine main body **24**. The continuous medium P on which an image has been formed can be wound from the side of the printing machine **12** by rotating the winding shaft **26** by the driving of the winding motor.

[0029] As illustrated in FIGS. **1** and **2**, the printing machine **12** includes a printing machine main body **30** having a housing shape, and an image former **32** that forms an image based on image data included in a print job is provided in an upper portion of the printing machine main body **30**. The image former **32** includes five image forming units **34W**, **34Y**, **34M**, **34C**, and **34K** for forming an image with color toners of W (white), Y (yellow), M (magenta), C (cyan), and K (black). The image forming units **34W**, **34Y**, **34M**, **34C**, and **34K** for W, Y, M, C, and K components are arranged along the up-down direction.

[0030] The image forming apparatus **10** forms an images with five toners of a white toner, a yellow toner, a magenta toner, a cyan toner, and a black toner. In other words, the image forming apparatus **10** forms an image with standard color toners of four colors (the yellow toner, the magenta toner, the cyan toner, and the black toner) and the white toner, which is a special toner other than the standard color toners of the four colors. Note that, the image forming apparatus **10** may form an image by using another special toner such as a clear toner instead of the white toner. The image forming apparatus **10** may form an image with the standard color toners of the four colors.

[0031] The image forming units **34W**, **34Y**, **34M**, **34C**, and **34K** for the W, Y, M, C, and K components have a similar configuration. For convenience of illustration and description, common constituent elements are denoted by the same reference signs, and in a case where the common constituent elements are distinguished from each other, a reference sign is indicated by adding W, Y, M, C, or K thereto. In FIG. **1**, reference signs are attached to only the constituent elements of the image forming units **34W** and **34Y** for the W and Y components, and reference signs are omitted for the constituent elements of the other image forming units **34M**, **34C**, and **34K**.

[0032] The image forming unit **34** includes a photoreceptor drum **36**, and includes an electric charger **38**, an exposer (scanning optical apparatus) **40**, a developer **42**, and a drum cleaner **44** which are disposed around the photoreceptor drum **36**.

[0033] The photoreceptor drum **36** is, for example, a negatively charged organic photo-conductor (OPC) in which an undercoat layer (UCL), a charge generation layer (CGL), and a charge transport layer (CTL) are sequentially laminated on a circumferential surface of a conductive cylindrical body made of aluminum. The photoreceptor drum **36** rotates at a constant circumferential speed by the driving of a rotation motor (not illustrated).

[0034] The electric charger **38** uniformly negatively charges the outer peripheral surface of the photoreceptor drum **36**. The exposer **40** emits laser light as scanning light to scan the outer circumferential surface of the photoreceptor drum **36** while exposing the outer circumferential surface of the photoreceptor drum **36**, thereby forming an electrostatic latent image on the outer

circumferential surface of the photoreceptor drum **36**. The amount of light of the laser light emitted from the exposers **40** is modulated according to image data in each color of WYMCK.

[0035] The developer **42** is, for example, a developer of a two-component developing system. The developer **42** visualizes the electrostatic latent image on the outer circumferential surface of the photoreceptor drum **36** by attaching the toner of each color component to the outer circumferential surface of the photoreceptor drum **36** to form a toner image. Further, the drum cleaner **44** cleans transfer residual toner remaining on the outer peripheral surface of the photoreceptor drum **36** after the primary transfer. The drum cleaner **44** includes a drum cleaning blade or the like that comes into sliding contact with the outer circumferential surface of the photoreceptor drum **36**.

[0036] As illustrated in FIGS. **1** and **2**, an intermediate transfer unit **46** for transferring a toner image to the continuous medium P is provided in the upper portion of the printing machine main body **30**. The intermediate transfer unit **46** includes an intermediate transfer belt **48** having an endless shape, a plurality of support rollers **50**, a primary transfer roller **52**, a secondary transfer roller **54**, and a belt cleaner **56**.

[0037] The intermediate transfer belt **48** extends in the up-down direction and is stretched around the plurality of support rollers **50**. The intermediate transfer belt **48** circulates and runs by the rotation of the plurality of support rollers **50**. At least one support roller **50** of the plurality of support rollers **50** is constituted by a driving roller coupled to a rotation motor (not illustrated), and the other support rollers **50** are constituted by a driven roller (free roller).

[0038] The primary transfer roller **52** is disposed on the side of the inner peripheral surface of the intermediate transfer belt **48** so as to face the photoreceptor drum **36** of each color component. The primary transfer roller **52** cooperates with the photoreceptor drum **36** to hold the intermediate transfer belt **48** therebetween. A primary transfer nip CN for transferring toner images from the photoreceptor drum **36** to the intermediate transfer belt **48** is formed between the outer peripheral surface of the primary transfer roller **52** and the outer peripheral surface of the photoreceptor drum **36**.

[0039] The secondary transfer roller **54** is disposed on the side of the outer peripheral surface of the intermediate transfer belt **48** so as to face a predetermined support roller **50**. The secondary transfer roller **54** cooperates with the predetermined support roller **50** to hold the intermediate transfer belt **48** therebetween. A secondary transfer nip TN for transferring toner images from the intermediate transfer belt **48** to the continuous medium P is formed between the outer peripheral surface of the secondary transfer roller **54** and the outer peripheral surface of the predetermined support roller **50**.

[0040] When the intermediate transfer belt **48** passes through the primary transfer nip CN, the toner images on the photoreceptor drums **36** are sequentially superimposed and primarily transferred onto the intermediate transfer belt **48**. Specifically, the toner images are electrostatically transferred onto the intermediate transfer belt **48** by applying primary transfer bias to the primary transfer roller **52**, and applying electric charge with a polarity opposite to that of the toner to the side of the rear surface of the intermediate transfer belt **48** (the side on which the intermediate transfer belt **48** abuts on the primary transfer roller **52**).

[0041] Thereafter, when the continuous medium P passes through the secondary transfer nip TN, the toner images on the intermediate transfer belt **48** are secondarily transferred to the continuous medium P. Specifically, the toner images are electrostatically transferred onto the continuous medium P by applying secondary transfer bias to the secondary transfer roller **54**, and applying electric charge with a polarity opposite to that of the toner to the side of the rear surface of the continuous medium P (the side on which the continuous medium P abuts on the secondary transfer roller **54**), and an image can be formed on the continuous medium P.

[0042] The belt cleaner **56** removes transfer residual toner remaining on the surface of the intermediate transfer belt **48** after the secondary transfer. The belt cleaner **56** includes a cleaning blade that comes into sliding contact with the surface of the intermediate transfer belt **48**.

[0043] As illustrated in FIGS. **1** and **3**, a fixer **58** that heats and fixes a toner image formed on the

continuous medium P is provided on the side of the exit of the secondary transfer nip TN in the printing machine main body **30**. The fixer **58** includes a heating roller **60**, a fixing roller **62**, a fixing belt **64** having an endless shape, a pressure roller **66**, and a temperature sensor **68**.

[0044] The heating roller **60** is disposed leftward from the intermediate transfer belt **48**. The heating roller **60** includes a hollow cylindrical core metal made of, for example, aluminum, iron, stainless steel (SUS), or the like, and a heating apparatus, such as a halogen lamp, for example, provided inside the core metal.

[0045] The fixing roller **62** is disposed at a position facing the heating roller **60**. The fixing roller **62** includes: a hollow cylindrical core metal made of, for example, aluminum, iron, SUS, or the like; an elastic layer **62e** provided on the outer circumferential surface of the core metal and made of, for example, silicone rubber or the like; and a surface layer laminated on the surface of the elastic layer **62e** and made of, for example, polytetrafluoroethylene (PTFE) or the like. The elastic layer **62e** of the fixing roller **62** is an elastic member that comes into indirect contact with the continuous medium P via the surface layer of the fixing roller **62** and the fixing belt **64**.

[0046] The fixing belt **64** is stretched around the heating roller **60** and the fixing roller **62**. The fixing belt **64** includes: a base material made of, for example, polyimide resin or the like; an elastic layer laminated on the surface of the base material and made of, for example, silicone rubber or the like; and a surface layer laminated on the surface of the elastic layer and made of, for example, polytetrafluoroethylene (PFA) or the like.

[0047] The pressure roller **66** is disposed at a position facing the fixing roller **62** outside the fixing belt **64**. The pressure roller **66** comes into pressure contact with the fixing roller **62** via the fixing belt **64** under a predetermined fixing load. The pressure roller **66** is configured to be able to come into pressure contact with and separate from the fixing roller **62**. The pressure roller **66** rotates by the driving of a rotation motor (not illustrated). The pressure roller **66** includes: a hollow cylindrical core metal made of, for example, aluminum, iron, SUS, or the like; an elastic layer **66e** provided on the outer peripheral surface of the core metal and made of, for example, silicone rubber or the like; and a surface layer provided on the surface of the elastic layer **66e** and made of, for example, a PFA tube or the like. The elastic layer **66e** of the pressure roller **66** is an elastic member that comes into indirect contact with the continuous medium P via the surface layer of the pressure roller **66**. The elastic layer **66e** of the pressure roller **66** may come into direct contact with the continuous medium P by omitting the surface layer of the pressure roller **66**.

[0048] A fixing nip FN for conveying the continuous medium P while heating and pressurizing the continuous medium P is formed between the fixing roller **62** and the pressure roller **66**. Further, the temperature sensor **68** is disposed at a position facing the heating roller **60**. The temperature sensor **68** detects the temperature of the heating roller **60** (fixing belt **64**).

[0049] When the pressure roller **66** rotates by the driving of a rotation motor, the fixing belt **64** is driven to circulate. As the fixing belt **64** circulates, the heating roller **60** and the fixing roller **62** are driven to rotate. Thus, the toner images can be fixed on the continuous medium P by conveying the continuous medium P while heating and pressurizing the continuous medium P at the fixing nip FN.

[0050] A plurality of conveyance roller pairs **70** for conveying the continuous medium P in the conveyance direction via the secondary transfer nip TN and the fixing nip FN is rotatably provided in the printing machine main body **30**.

[0051] In order to remove ultra fine particles (UFP) generated from the elastic layer **62e** of the fixing roller **62** and the elastic layer **66e** of the pressure roller **66**, the image forming apparatus **10** adopts the following configuration. The ultra fine particles are, for example, metal or ceramic particles having a diameter of 0.1 μm or less, and each particle is an aggregate of several hundred or more atoms.

[0052] As illustrated in FIGS. **3** and **4**, a first opening **72** having a rectangular shape is formed in a wall portion on the side of the rear surface of the printing machine main body **30**. As a first

collector that collects ultra fine particles, a first filter **74** is provided on the side of the first opening **72** in the wall portion on the side of the rear surface of the printing machine main body **30**. The first filter **74** covers the first opening **72**, and is configured to be replaceably attachable to and detachable from the wall portion on the side of the rear surface of the printing machine main body **30**.

[0053] On the side of the exit of the first filter **74** in the wall portion on the rear side of the printing machine main body **30**, a plurality of first air blowing fans **76** as a first air blower that generates an air current toward the side of the first filter **74** around the fixer **58** is provided via a bracket or the like. The plurality of first air blowing fans **76** sucks air around the fixer **58** to the side of the first filter **74**. In other words, the plurality of first air blowing fans **76** sucks ultra fine particles included in air around the fixer **58** to the side of the first filter **74**. The output mode of the plurality of first air blowing fans **76** as the first air blower is configured to be switchable among a low output mode, a first high output mode, and a second high output mode. The output mode of the first air blower can be switched by changing the number of first air blowing fans **76** to be driven or the rotation speed of the first air blowing fans **76**.

[0054] The low output mode in the first air blower (the plurality of first air blowing fans **76**) is an output mode of 40% or less (for example, 30%) of the maximum output of the plurality of first air blowing fans **76**. The first high output mode in the first air blower is an output mode higher than the low output mode, and is an output mode of 70% or more (for example, 100%) of the maximum output of the plurality of first air blowing fans **76**. The second high output mode in the first air blower is an output mode lower than the first high output mode and is an output mode of 70% or more (for example, 80%) of the maximum output of the plurality of first air blowing fans **76**.

[0055] As illustrated in FIGS. **1**, **2**, and **5**, a second opening **78** having a rectangular shape is formed at a position corresponding to each of the image forming units **34W**, **34Y**, **34M**, **34C**, and **34K** in a wall portion on the side of the front surface of the printing machine main body **30**. A second filter **80** as a second collector that collects ultra fine particles is provided on the side of each second opening **78** in the wall portion on the side of the front surface of the printing machine main body **30**. Each second filter **80** covers each second opening **78**, and is configured to be replaceably attachable to and detachable from the wall portion on the side of the front surface of the printing machine main body **30**.

[0056] On the side of the exit of each second filter **80** in the wall portion on the side of the front surface of the printing machine main body **30**, a second air blowing fan **82** as a second air blower that generates an air current toward the side of each second filter **80** around each image forming unit **34** is provided via a bracket or the like. Each second air blowing fan **82** sucks air around each image forming unit **34** to the side of the second filter **80**. In other words, each second air blowing fan **82** sucks ultra fine particles included in air around each image forming unit **34** to the side of the second filter **80**. Each second air blowing fan **82** ejects air (clean air) cleaned by each second filter **80**. The output mode of each second air blowing fan **82** as the second air blower is configured to be switchable among a low output mode, a first high output mode, and a second high output mode. The output mode of the second air blower can be switched by changing the rotation speed of each second air blowing fan **82**.

[0057] The low output mode in the second air blower (the plurality of second air blowing fans **82**) is an output mode of 40% or less (for example, 30%) of the maximum output of the plurality of second air blowing fans **82**. The first high output mode in the second air blower is an output mode higher than the low output mode, and is an output mode of 70% or more (for example, 100%) of the maximum output of the plurality of second blowing fans **82**. The second high output mode in the second blowing part is an output mode lower than the first high output mode, and is an output mode of 70% or more (for example, 80%) of the maximum output of the plurality of second blowing fans **82**.

[0058] As illustrated in FIGS. **2** and **5**, a supply duct **84** that supplies the clean air ejected from

each second air blowing fan **82** to a charged electrode of each electric charger **38** is provided below each developer **42**, and each supply duct **84** extends in the front-rear direction along each electric charger **38**. Each supply duct **84** is connected to the ejection side of each second air blowing fan **82** via a communication pipe **86**.

[0059] Either the first filter **74** as the first collector and the plurality of second filters **80** as the second collector may be omitted from the configuration of the image forming apparatus **10**. In a case where the first filter **74** is omitted, the plurality of first air blowing fans **76** as the first air blower is omitted. In a case where the plurality of second filters **80** is omitted, the plurality of second blowing fans **82** as the second air blower, the plurality of supply ducts **84**, and the plurality of communication pipes **86** are omitted.

[0060] A control configuration of the image forming apparatus **10** according to the present embodiment will be described with reference to FIG. **6**. FIG. **6** is a control block diagram of the image forming apparatus **10** according to the present embodiment.

[0061] As illustrated in FIG. **6**, the image forming apparatus **10** includes a controller **88** that controls the image former **32**, the intermediate transfer unit **46**, the fixer **58**, the plurality of first air blowing fans **76**, the plurality of second air blowing fans **82**, and the like. The controller **88** plays a role in managing control of the image forming apparatus **10** in its entirety. The controller **88** includes a central processing unit (CPU) **90**, a random access memory (RAM) **92**, and a read only memory (ROM) **94**. A storage **96** and a communicator **98** are connected to the controller **88**.

[0062] The CPU **90** comprehensively controls the entire operation of the image forming apparatus **10**. The CPU **90** reads various control programs and setting data stored in the ROM **92**, develops the read control programs and setting data in the RAM **94**, and executes the programs to perform various pieces of arithmetic processing. The RAM **94** provides a working memory space for the CPU **90** and stores temporary data. Further, when the CPU **90** performs various pieces of arithmetic processing, the CPU **90** refers to various kinds of data stored in the storage **96**. The storage **96** is configured by, for example, a non-volatile semiconductor memory or a hard disk drive.

[0063] The controller **88** transmits and receives various kinds of data to and from an external apparatus (for example, a personal computer) connected to a communication network such as a local area network (LAN) or a wide area network (WAN) via the communicator **98**. The controller **88** receives, for example, a print job transmitted from the external apparatus, and causes a toner image to be formed on the continuous medium **P** based on image data (input image data) included in the print job. The communicator **98** is constituted by, for example, a communication control card such as a LAN card.

[0064] The temperature sensor **68** is connected to the controller **88**. The controller **88** monitors the temperature of the heating roller **60** in the fixer **58**. In addition, the controller **88** (the CPU **90**) has the following configuration by executing various control programs.

[0065] As illustrated in FIG. **6**, the controller **88** varies the output mode of the plurality of first air blowing fans **76** as the first air blower and the plurality of second blowing fans **82** as the second air blower according to at least one of an apparatus state of the image forming apparatus **10** and/or a printing condition (image forming condition) for the image forming apparatus **10**. Further, the controller **88** may vary the output mode of only either the plurality of first air blowing fans **76** or the plurality of second air blowing fans **82** according to the apparatus state of the image forming apparatus **10** or the like.

[0066] The apparatus state of the image forming apparatus **10** includes whether image formation is in progress (printing is in progress). In a case where image formation is in progress as the apparatus state of the image forming apparatus **10**, the controller **88** sets the output mode of the plurality of first air blowing fans **76** as the first air blower and the second blowing fan **82** as the second air blower to a high output mode. The high output mode includes the first high output mode and the second high output mode as described above. In a case where image formation is in progress as the apparatus state of the image forming apparatus **10**, the controller **88** sets the output mode of the

plurality of first air blowing fans **76** and the plurality of second air blowing fans **82** to one of the first high output mode and the second high output mode. In a case where image formation is not in progress as the apparatus state of the image forming apparatus **10**, the controller **88** sets the output mode of the plurality of first air blowing fans **76** and the plurality of second air blowing fans **82** to the low output mode. The case where image formation is not in progress as the apparatus state of the image forming apparatus **10** means a case of being idling.

[0067] As illustrated in FIGS. **3** and **6**, the apparatus state of the image forming apparatus **10** includes the temperature of the heating roller **60** in the fixer **58**. In a case where the temperature of the heating roller **60** is equal to or higher than a predetermined temperature, the controller **88** may set the output mode of the plurality of first air blowing fans **76** and the plurality of second air blowing fans **82** to one of the first high output mode and the second high output mode. In a case where the temperature of the heating roller **60** is less than the predetermined temperature, the controller **88** may set the output mode of the plurality of first air blowing fans **76** and the plurality of second air blowing fans **82** to the low output mode. The predetermined temperature is a temperature serving as a reference for switching the output mode of the plurality of first air blowing fans **76** and the plurality of second air blowing fans **82** to the high output mode, and is, for example, 195° C. in the present embodiment. Note that, whether the temperature of the heating roller **60** is equal to or higher than the predetermined temperature corresponds to whether image formation is in progress as the apparatus state of the image forming apparatus **10**.

[0068] The apparatus state of the image forming apparatus **10** includes whether the fixing roller **62** and the pressure roller **66** in the fixer **58** are separated from each other. In a case where the fixing roller **62** and the pressure roller **66** are separated from each other, the controller **88** may set the output mode of the plurality of first air blowing fans **76** and the plurality of second blowing fans **82** to the low output mode. In a case where the fixing roller **62** and the pressure roller **66** are not separated from each other, the controller **88** may set the output mode of the plurality of first air blowing fans **76** and the plurality of second air blowing fans **82** to one of the first high output mode and the second high output mode. Note that, whether the fixing roller **62** and the pressure roller **66** are separated from each other corresponds to whether it is idling.

[0069] As illustrated in FIG. **6**, a plurality of the printing conditions includes a printing distance (image forming distance) for forming an image along the longitudinal direction of the continuous medium **P**. The controller **88** varies the output mode of the plurality of first air blowing fans **76** and the plurality of second air blowing fans **82** according to the printing distance under a predetermined condition. The controller **88** may set the output mode of the plurality of first air blowing fans **76** and the plurality of second air blowing fans **82** to the first high output mode in a case where the printing distance is equal to or greater than a predetermined distance under the predetermined condition. The controller **88** temporarily sets the output mode of the plurality of first air blowing fans **76** and the plurality of second air blowing fans **82** to the second high output mode in a case where the printing distance is less than the predetermined distance under the predetermined condition. The predetermined condition means that image formation is in progress as the apparatus state of the image forming apparatus **10**, or refers to a corresponding state corresponding thereto. The corresponding state means that the temperature of the heating roller **60** is equal to or higher than the predetermined temperature, or that the fixing roller **62** and the pressure roller **66** are not separated from each other. The predetermined distance means a distance serving as a reference for switching the output mode of the plurality of first air blowing fans **76** and the plurality of second air blowing fans **82** to the first high output mode, and is, for example, 300 m in the present embodiment.

[0070] The plurality of printing conditions includes whether toners of five or more colors are used for image formation (printing). The controller **88** may vary the output mode of the plurality of first air blowing fans **76** and the plurality of second air blowing fans **82** according to whether the toners of five or more colors are used for the image formation under the predetermined condition. The

controller **88** may set the output mode of the plurality of first air blowing fans **76** and the plurality of second air blowing fans **82** to the first high output mode in a case where the toners of five or more colors are used for the image formation under the predetermined condition. In a case where the toners of five or more colors are not used for the image formation under the predetermined condition, the controller **88** may set the output mode of the plurality of first air blowing fans **76** and the plurality of second air blowing fans **82** to the first high output mode.

[0071] The plurality of printing conditions includes the toner adhesion amount per unit area of the continuous medium P. The controller **88** varies the output mode of the plurality of first air blowing fans **76** and the plurality of second air blowing fans **82** according to the toner adhesion amount per unit area of the continuous medium P under the predetermined condition. The controller **88** sets the output mode of the plurality of first air blowing fans **76** and the plurality of second air blowing fans **82** to the first high output mode in a case where the toner adhesion amount per unit area of the continuous medium P is equal to or greater than a predetermined adhesion amount under the predetermined condition. The controller **88** may set the output mode of the plurality of first air blowing fans **76** and the plurality of second air blowing fans **82** to the second high output mode in a case where the toner adhesion amount per unit area of the continuous medium P is less than the predetermined adhesion amount under the predetermined condition. The predetermined adhesion amount is an adhesion amount serving as a reference for switching the output mode of the plurality of first air blowing fans **76** and the plurality of second air blowing fans **82** to the first high output mode, and is, for example, an adhesion amount equal to or greater than half the maximum adhesion amount per unit area of the continuous medium P in the present embodiment.

[0072] The plurality of printing conditions includes whether a special toner such as the white toner is used for image formation. The controller **88** varies the output mode of the plurality of first air blowing fans **76** and the plurality of second air blowing fans **82** according to whether a special toner is used for image formation under the predetermined condition. The controller **88** sets the output mode of the plurality of first air blowing fans **76** and the plurality of second air blowing fans **82** to the first high output mode in a case where a special toner is used for the image formation under the predetermined condition. The controller **88** may set the output mode of the plurality of first air blowing fans **76** and the plurality of second air blowing fans **82** to the second high output mode in a case where a special toner is not used for the image formation under the predetermined condition. The controller **88** may set the output mode of the plurality of first air blowing fans **76** and the plurality of second air blowing fans **82** to the first high output mode in a case where a special toner is not used for the image formation under the predetermined condition.

[0073] The plurality of printing conditions includes whether the continuous medium P is a special medium such as tack paper or a PET film. The controller **88** varies the output mode of the plurality of first air blowing fans **76** and the plurality of second blowing fans **82** according to whether the continuous medium P is the special medium under the predetermined condition. The controller **88** may set the output mode of the plurality of first air blowing fans **76** and the plurality of second blowing fans **82** to the first high output mode in a case where the continuous medium P is the special medium under the predetermined condition. The controller **88** temporarily sets the output mode of the plurality of first air blowing fans **76** and the plurality of second blowing fans **82** to the second high output mode in a case where the continuous medium P is not the special medium under the predetermined condition.

[0074] The plurality of printing conditions includes the thickness of the continuous medium P. The controller **88** may vary the output mode of the plurality of first air blowing fans **76** and the plurality of second air blowing fans **82** according to the thickness of the continuous medium P under the predetermined condition. The controller **88** may set the output mode of the plurality of first air blowing fans **76** and the plurality of second blowing fans **82** to the first high output mode in a case where the thickness of the continuous medium P is equal to or greater than a predetermined thickness under the predetermined condition. The controller **88** may set the output mode of the

plurality of first air blowing fans **76** and the plurality of second air blowing fans **82** to the second high output mode in a case where the thickness of the continuous medium **P** is less than the predetermined thickness under the predetermined condition. The predetermined thickness is a thickness serving as a reference for switching the output mode of the plurality of first air blowing fans **76** and the plurality of second air blowing fans **82** to the first high output mode, and is, for example, 200 μm in the present embodiment.

[0075] Setting processing of the output mode of the plurality of first air blowing fans **76** and the plurality of second air blowing fans **82** will be described with reference to FIG. 7. FIG. 7 is a flowchart for describing the setting processing of the output mode of the plurality of first air blowing fans **76** and the plurality of second air blowing fans **82**.

[0076] The controller **88** determines whether image formation is in progress as the apparatus state of the image forming apparatus **10** (step **S101** in FIG. 7). In a case where image formation is not in progress as the apparatus state of the image forming apparatus **10** (NO in step **S101** in FIG. 7), the controller **88** sets the plurality of first air blowing fans **76** and the plurality of second air blowing fans **82** to the low output mode (step **S102** in FIG. 7).

[0077] In a case where image formation is in progress as the apparatus state of the image forming apparatus **10** (YES in step **S101** in FIG. 7), the controller **88** determines whether the printing distance is equal to or greater than the predetermined distance (step **S103** in FIG. 7). In a case where the printing distance is less than the predetermined distance (NO in step **S103** in FIG. 7), the controller **88** temporarily sets the plurality of first air blowing fans **76** and the plurality of second air blowing fans **82** to the second high output mode (step **S104** in FIG. 7).

[0078] In a case where the printing distance is equal to or greater than the predetermined distance (YES in step **S103** in FIG. 7), the controller **88** determines whether the continuous medium **P** is the special medium (step **S105** in FIG. 7). In a case where the continuous medium **P** is not the special medium (NO in step **S105** in FIG. 7), the controller **88** temporarily sets the output mode of the plurality of first air blowing fans **76** and the plurality of second air blowing fans **82** to the second high output mode (step **S106** in FIG. 7). In a case where the continuous medium **P** is the special medium (YES in step **S105** in FIG. 7), the controller **88** determines whether the toner adhesion amount per unit area of the continuous medium **P** is equal to or greater than the predetermined adhesion amount (step **S107** in FIG. 7). In a case where the toner adhesion amount is equal to or greater than the predetermined adhesion amount (YES in step **S107** in FIG. 7), the controller **88** sets the output mode of the plurality of first air blowing fans **76** and the plurality of second air blowing fans **82** to the first high output mode (step **S108** in FIG. 7).

[0079] In a case where the toner adhesion amount is less than the predetermined adhesion amount (NO in step **S107** in FIG. 7), the controller **88** determines whether a special toner is used for the image formation (step **S109** in FIG. 7). In a case where a special toner is used (YES in step **S109** in FIG. 7), the controller **88** sets the output mode of the plurality of first air blowing fans **76** and the plurality of second air blowing fans **82** to the first high output mode (step **S108** in FIG. 7).

[0080] In a case where a special toner is not used (NO in step **S109** in FIG. 7), the controller **88** determines whether the output mode of the plurality of first air blowing fans **76** and the plurality of second air blowing fans **82** has been temporarily set to the second high output mode (step **S110** in FIG. 7). In a case where the second high output mode has been temporarily set (YES in step **S110** in FIG. 7), the controller **88** sets the output mode of the plurality of first air blowing fans **76** and the plurality of second air blowing fans **82** to the second high output mode (step **S111** in FIG. 7). In a case where the second high output mode has not been temporarily set (NO in step **S110** in FIG. 7), the controller **88** sets the output mode of the plurality of first air blowing fans **76** and the plurality of second air blowing fans **82** to the first high output mode (step **S108** in FIG. 7).

[0081] Note that, the setting processing of the output mode of the plurality of first air blowing fans **76** and the plurality of second air blowing fans **82** described above is an example, and may be changed as appropriate.

[0082] According to the configuration of the image forming apparatus **10** according to the present embodiment, ultra fine particles included in air around the fixer **58** are collected by the first filter **74** by driving the plurality of first air blowing fans **76** to generate an air current toward the side of the first filter **74** around the fixer **58**. In addition, ultra fine particles included in air around each image forming unit **34** are collected by each second filter **80** by driving each second air blowing fan **82** to generate an air current toward the side of each second filter **80** around each image forming unit **34**. Accordingly, the present embodiment makes it possible to form a high-quality image on the continuous medium P by removing ultra fine particles which cause image streaks and image unevenness.

[0083] According to the configuration of the image forming apparatus **10** according to the present embodiment, the controller **88** varies the output mode of the plurality of first air blowing fans **76** and the plurality of second blowing fans **82** according to at least one of the apparatus state of the image forming apparatus **10** and/or the printing condition for the image forming apparatus **10**. For this reason, the output of the plurality of first air blowing fans **76** and the plurality of second air blowing fans **82** does not become excessively high and is set (adjusted) to an output corresponding to at least one of the apparatus state of the image forming apparatus **10** and/or the printing condition for the image forming apparatus **10**. Accordingly, the present embodiment makes it possible to attempt a noise reduction in the image forming apparatus **10** by suppressing the fan noise of the plurality of first air blowing fans **76** and the like as well as attempt energy saving for the image forming apparatus **10** by suppressing the power consumption of the plurality of first air blowing fans **76** and the like.

[0084] In addition, according to the configuration of the image forming apparatus **10** according to the present embodiment, the controller **88** sets the output mode of the plurality of first air blowing fans **76** and the like to the low output mode in a case where image formation is not in progress as the apparatus state of the image forming apparatus **10**. For this reason, the present embodiment makes it possible to sufficiently suppress the fan noise and the power consumption of the plurality of first air blowing fans **76** and the like to promote a noise reduction in and energy saving for the image forming apparatus **10** in a case where image formation is not in progress as the apparatus state of the image forming apparatus **10**.

[0085] According to the configuration of the image forming apparatus **10** according to the present embodiment, the controller **88** sets the output mode of the plurality of first air blowing fans **76** and the like to the low output mode in a case where the temperature of the heating roller **60** is less than the predetermined temperature. For this reason, the present embodiment makes it possible to sufficiently suppress the fan noise and the power consumption of the plurality of first air blowing fans **76** and the like to promote a noise reduction in and energy saving for the image forming apparatus **10** in a case where the temperature of the heating roller **60** is less than the predetermined temperature.

[0086] According to the configuration of the image forming apparatus **10** according to the present embodiment, the controller **88** sets the output mode of the plurality of first air blowing fans **76** and the plurality of second air blowing fans **82** to the low output mode in a case where the fixing roller **62** and the pressure roller **66** are separated from each other. For this reason, the present embodiment makes it possible to sufficiently suppress the fan noise and the power consumption of the plurality of first air blowing fans **76** and the like to promote a noise reduction in and energy saving for the image forming apparatus **10** in a case where the fixing roller **62** and the pressure roller **66** are separated from each other.

[0087] In addition, according to the configuration of the image forming apparatus **10** according to the present embodiment, the controller **88** varies the output mode of the plurality of first air blowing fans **76** and the like according to the printing distance. For this reason, the present embodiment makes it possible to sufficiently suppress the fan noise and the power consumption of the plurality of first air blowing fans **76** and the like to promote a noise reduction in and energy

saving for the image forming apparatus **10** in a case where the printing distance is less than the predetermined distance.

[0088] According to the configuration of the image forming apparatus **10** according to the present embodiment, the controller **88** varies the output mode of the plurality of first air blowing fans **76** and the like according to whether the toners of five or more colors are used for the image formation under the predetermined condition. For this reason, the present embodiment makes it possible to sufficiently suppress the fan noise and the power consumption of the plurality of first air blowing fans **76** and the like to promote a noise reduction in and energy saving for the image forming apparatus **10** in a case where the toners of five or more colors are not used for the image formation.

[0089] According to the configuration of the image forming apparatus **10** according to the present embodiment, the controller **88** varies the output mode of the plurality of first air blowing fans **76** and the like according to the toner adhesion amount per unit area of the continuous medium P. For this reason, the present embodiment makes it possible to sufficiently suppress the fan noise and the power consumption of the plurality of first air blowing fans **76** and the like to promote a noise reduction in and energy saving for the image forming apparatus **10** in a case where the toner adhesion amount is less than the predetermined adhesion amount.

[0090] According to the configuration of the image forming apparatus **10** according to the present embodiment, the controller **88** varies the output mode of the plurality of first air blowing fans **76** and the like according to whether a special toner is used for the image formation. For this reason, the present embodiment makes it possible to sufficiently suppress the fan noise and the power consumption of the plurality of first air blowing fans **76** and the like to promote a noise reduction in and energy saving for the image forming apparatus **10** in a case where a special toner is not used.

[0091] According to the configuration of the image forming apparatus **10** according to the present embodiment, the controller **88** varies the output mode of the plurality of first air blowing fans **76** and the like according to whether the continuous medium P is the special medium. For this reason, the present embodiment makes it possible to sufficiently suppress the fan noise and the power consumption of the plurality of first air blowing fans **76** and the like to promote a noise reduction in and energy saving for the image forming apparatus **10** in a case where the continuous medium P is not the special medium.

[0092] According to the configuration of the image forming apparatus **10** according to the present embodiment, the controller **88** varies the output mode of the plurality of first air blowing fans **76** and the like according to the thickness of the continuous medium P. For this reason, the present embodiment makes it possible to sufficiently suppress the fan noise and the power consumption of the plurality of first air blowing fans **76** and the like to promote a noise reduction in and energy saving for the image forming apparatus **10** in a case where the thickness of the continuous medium P is less than the predetermined thickness.

[0093] Although the present embodiment has been specifically described above, the present invention is not limited to the above-described specific embodiment. Various modifications and changes can be made to the specific examples described in the above embodiment within the scope of the spirit of the present invention described in the claims.

[0094] Although embodiments of the present invention have been described and illustrated in detail, it is clearly understood that the same is by way of illustration and example only and not limitation, the scope of the present invention should be interpreted by terms of the appended claims.

Claims

1. An image forming apparatus that forms an image on a continuous medium, the image forming apparatus comprising: a fixer that includes an elastic member and heats and fixes a toner image formed on the continuous medium, the elastic member coming into in direct or indirect contact

- with the continuous medium; a collector that collects ultra fine particles generated from the elastic member; an air blower that generates an air current toward a side of the collector; and a hardware processor that varies an output mode of the air blower according to at least one of an apparatus state of the image forming apparatus and/or a printing condition for the image forming apparatus.
2. The image forming apparatus according to claim 1, wherein: the apparatus state includes whether image formation is in progress, and the hardware processor is sets, in a case where the image formation is in progress as the apparatus state, the output mode of the air blower to a high output mode, and sets, in a case where the image formation is not in progress as the apparatus state, the output mode of the air blower to a low output mode.
 3. The image forming apparatus according to claim 1, wherein: the apparatus state includes a temperature of the fixer, and the hardware processor sets, in a case where the temperature of the fixer is equal to or higher than a predetermined temperature, the output mode of the air blower to a high output mode, and sets, in a case where the temperature of the fixer is less than the predetermined temperature, the output mode of the air blower to a low output mode, the predetermined temperature serving as a reference for switching the output mode of the air blower to the high output mode.
 4. The image forming apparatus according to claim 1, wherein: a plurality of the printing conditions includes a printing distance for forming the image along a longitudinal direction of the continuous medium, and the hardware processor varies the output mode of the air blower according to the printing distance.
 5. The image forming apparatus according to claim 1, wherein: a plurality of the printing conditions includes whether toners of five or more colors are used for image formation, and the hardware processor varies the output mode of the air blower according to whether the toners of the five or more colors are used for the image formation.
 6. The image forming apparatus according to claim 1, wherein: a plurality of the printing conditions includes a toner adhesion amount per unit area of the continuous medium, and the hardware processor varies the output mode of the air blower according to the toner adhesion amount.
 7. The image forming apparatus according to claim 1, wherein: a plurality of the printing conditions includes whether a special toner is used for image formation, and the hardware processor varies the output mode of the air blower according to whether the special toner is used for the image formation.
 8. The image forming apparatus according to claim 7, wherein the special toner is a white toner.
 9. The image forming apparatus according to claim 1, wherein: a plurality of the printing conditions includes whether the continuous medium is a special medium, and the hardware processor varies the output mode of the air blower according to whether the continuous medium is the special medium.
 10. The image forming apparatus according to claim 9, wherein the special medium is tack paper.
 11. The image forming apparatus according to claim 1, wherein: a plurality of the printing conditions includes a thickness of the continuous medium, and the hardware processor varies the output mode of the air blower according to the thickness of the continuous medium.
 12. The image forming apparatus according to claim 1, wherein: the apparatus state includes whether a fixing roller and a pressure roller in the fixer are separated from each other, and the hardware processor sets, in a case where the fixing roller and the pressure roller are separated from each other, the output mode of the air blower to a low output mode, and sets, in a case where the fixing roller and the pressure roller are not separated from each other, the output mode of the air blower to a high output mode.
 13. The image forming apparatus according to claim 2, wherein the high output mode is an output mode in which an output of the air blower is 70% or more of a maximum output of the air blower,

and the low output mode is an output mode in which the output of the air blower is 40% or less of the maximum output of the air blower.
